

Technology Stack

Date	15 July 2026
Team ID	SWTID-2026-1518
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML, CSS, JavaScript	Provides a responsive interface for entering applicant details and displaying prediction results.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python (Flask)	Handles application logic and integrates the machine learning prediction model.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	MySQL	Stores applicant information, prediction history, and user data securely.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Docker	Provides consistent deployment across development and production environments.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub	Supports version control and team collaboration.

6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Scikit-learn, Pandas, NumPy, Jupyter Notebook	Used for data preprocessing, model training, evaluation, and credit card approval prediction.
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